COMPATIBILITY OF THE GYSIN MAP WITH THE TWO SECTIONS OF THE DECOMPOSED ZERO SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The decomposition of the zero scheme of a section of a tensor product of invertible sheaves. Same chapter chow.tex, label lemma-decompose-section, invoked in the statement at proof line 5885 and used throughout. Supplies the commutative diagram with p : L -> Y, the two sections s_i, the line bundles D_i -> D and the identification of D_1 with the relative spectrum of the inverse of L_2; the compatibility computation itself is in the counted span.
- Additivity of capping with a first Chern class and compatibility with flat pullback. Same chapter chow.tex, labels lemma-c1-cap-additive (proof line 5900) and lemma-flat-pullback-cap-c1 (proof line 5925). Gives i^*[W] = c_1(L_1) cap [W] + c_1(L_2) cap [W] when W is contained in D, and allows c_1 to be moved across the flat pullback p_1^* in the chain of equalities.
- Formulae for a line bundle and its zero section. Same chapter chow.tex, label lemma-linebundle-formulae, invoked at proof line 5927. Identifies the class of the zero section W_1 intersect W_2 of the line bundle W_1 -> W with p_1^*(c_1(L_2) cap [W]), which is the step that closes the first case.

Source reference. Original lemma 02SP: sources/stacks-chow/pinned-original-source.tex, source lines 4330–4330. Navigation pointer for \ref{lemma-c1-cap-additive} used inside the counted span (line 5900); the argument that consumes it is retained in the span.

Source reference. Original lemma 0F97: sources/stacks-chow/pinned-original-source.tex, source lines 5822–5822. Navigation pointer for \ref{lemma-decompose-section}

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used inside the counted span (line 5885); the argument that consumes it is retained in the span.

Source reference. Original lemma 02SS: sources/stacks-chow/pinned-original-source.tex, source lines 4533–4533. Navigation pointer for \ref{lemma-flat-pullback-cap-c1} used inside the counted span (line 5925); the argument that consumes it is retained in the span.

Source reference. Original lemma 0F96: sources/stacks-chow/pinned-original-source.tex, source lines 5775–5775. Navigation pointer for \ref{lemma-linebundle-formulae} used inside the counted span (line 5927); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 5887); the argument that consumes it is retained in the span.

Lemma 0.1. *In the situation of Lemma 0F97 assume Y is locally of finite type over (S, δ) as in Situation 02QL. Then we have $i_1^* p^* \alpha = p_1^* i^* \alpha$ in $\mathrm{CH}_k(D_1)$ for all $\alpha \in \mathrm{CH}_k(Y)$.*

Proof. Let $W \subset Y$ be an integral closed subscheme of δ-dimension k. We distinguish two cases.

Assume $W \subset D$. Then $i^*[W] = c_1(\mathcal{L}_1) \cap [W] + c_1(\mathcal{L}_2) \cap [W]$ in $\mathrm{CH}_{k-1}(D)$ by our definition of Gysin homomorphisms and the additivity of Lemma 02SP. Hence $p_1^* i^*[W] = p_1^*(c_1(\mathcal{L}_1) \cap [W]) + p_1^*(c_1(\mathcal{L}_2) \cap [W])$. On the other hand, we have $p^*[W] = [p^{-1}(W)]_{k+1}$ by construction of flat pullback. And $p^{-1}(W) = W_1 \cup W_2$ (scheme theoretically) where $W_i = p_i^{-1}(W)$ is a line bundle over W by the lemma (since formation of the diagram commutes with base change). Then $[p^{-1}(W)]_{k+1} = [W_1] + [W_2]$ as W_i are integral closed subschemes of L of δ-dimension $k+1$. Hence

$$\begin{aligned} i_1^* p^*[W] &= i_1^*[p^{-1}(W)]_{k+1} \\ &= i_1^*([W_1] + [W_2]) \\ &= c_1(p_1^*\mathcal{L}_1) \cap [W_1] + [W_1 \cap W_2]_k \\ &= c_1(p_1^*\mathcal{L}_1) \cap p_1^*[W] + [W_1 \cap W_2]_k \\ &= p_1^*(c_1(\mathcal{L}_1) \cap [W]) + [W_1 \cap W_2]_k \end{aligned}$$

by construction of Gysin homomorphisms, the definition of flat pullback (for the second equality), and compatibility of $c_1 \cap -$ with flat pullback (Lemma 02SS). Since $W_1 \cap W_2$ is the zero section of the line bundle $W_1 \to W$ we see from Lemma 0F96 that $[W_1 \cap W_2]_k = p_1^*(c_1(\mathcal{L}_2) \cap [W])$. Note that here we use the fact that D_1 is the line bundle which is the relative spectrum of the inverse of $\mathcal{L}_2$. Thus we get the same thing as before.

Assume $W \not\subset D$. In this case, both $i_1^* p^*[W]$ and $p_1^* i^*[W]$ are represented by the $k-1$ cycle associated to the scheme theoretic inverse image of W in D_1. $\square$